ARCHANA RODDA

Data Engineer

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Summary

Experienced Data Engineer with over 3 years in designing and managing large-scale data pipelines in the healthcare and finance sectors. Proficient in big data technologies such as Hadoop, Apache Spark, Kafka, and Snowflake, along with cloud platforms like Azure and AWS. Skilled in Python, PySpark, and SQL for data management and analysis. Adept at building ETL/ELT pipelines, enhancing data quality, and delivering insights through Power BI dashboards. Strong problem-solving skills with a proven track record of improving processing efficiency and supporting real-time analytics. Passionate about leveraging data to drive insights and efficiency, with certification in Azure Data Engineering and a solid background in agile methodologies.

Professional Experience

Blue Cross Blue Shield, PA **Aug 2024 - Current**

Data Engineer

- Developed and maintained a data pipeline to extract data from on premise SQL Server and load it into Amazon Redshift using AWS Lambda, S3, and Glue for ETL processing.
- Automated data ingestion processes from SQL Server to S3, ensuring data extraction and transformation using server less AWS components like Lambda, reducing manual data handling by 90%.
- Used PySpark and Spark SQL within AWS Glue to perform complex data transformations, cleaning, and schema normalization, converting raw data into optimized formats like Parquet for faster query performance in Redshift.
- Designed and optimized Redshift tables by implementing distribution and sort keys, improving query performance for large datasets by 40%.
- Orchestrated ETL workflows using Apache Airflow, scheduling jobs to run daily for continuous data processing, resulting in a fully automated batch-processing pipeline.
- Configured Amazon S3 as the central data lake, managing raw and processed data efficiently while securing access through IAM roles and bucket policies.
- Monitored pipeline performance using AWS Cloud Watch, setting up alerts to track job failures and execution times, improving overall system reliability by 30%.
- Collaborated with cross-functional teams to gather requirements, ensuring the data pipeline met business needs and supported advanced analytics on healthcare claims data.

ACS Solutions, India **Dec 2019 – Jul 2022**

Data Engineer

- Implemented large-scale data ingestion solutions from various sources, including SQL Server databases, Azure Blob Storage, IoT devices, and third-party API services, managing over 2 million records monthly.
- Developed real-time data pipelines using Azure Event Hubs and Kafka, processing telemetry data from over 50,000 patients per hour, significantly enhancing monitoring capabilities for healthcare providers.
- Processed raw data from Azure Data Lake Storage and SQL databases into Spark DataFrames, performing essential transformations to facilitate data reporting and analytics workflows.
- Migrated and optimized over 100 complex Hive/SQL queries into Spark transformations using Spark RDD, Scala, and Python, resulting in a 25% increase in processing efficiency.
- Built efficient ETL pipelines on Azure Databricks, ingesting and processing data from multiple sources each month to ensure data was transformed and prepared for analysis.
- Developed and deployed Azure Functions to aggregate data from various Event Hubs, securely storing processed data in Azure Cosmos DB for scalable access.
- Demonstrated expertise in Microsoft Azure Services, managing resources across Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics, Azure Data Factory, HDInsight, and Power BI for comprehensive data processing.
- Created and maintained interactive dashboards in Power BI, visualizing sales and healthcare data to improve organizational decision-making.
- Partnered with data analysts to implement DBT models, transforming raw data into structured, business-ready datasets for comprehensive reporting and analysis.
- Applied data modeling techniques in data warehousing, utilizing snowflake schemas to manage data models for optimized query performance and effective data organization.
- Scheduled, monitored, and debugged Azure Data Factory pipelines, ensuring the seamless flow of data between systems and maintaining data integrity.
- Optimized complex SQL queries to resolve data discrepancies reported by business users, ensuring accuracy and reliability of data, and improving overall data quality.
- Built machine-learning models using PyTorch and Scikit-Learn to predict patient outcomes, contributing to a 15% improvement in patient care through data-driven insights.

- Worked in an Agile/SCRUM environment, delivering new features and updates every two weeks while ensuring continuous integration of features without disrupting healthcare services.
- Utilized Azure DevOps for continuous integration and deployment, streamlining development workflows and ensuring efficient management of the development lifecycle.

KPMG, India **Apr 2019 – Nov 2019**

Data Analyst

- Assisted in data analysis projects focused on assessing client financial performance by collecting and analyzing data from various sources, including accounting software and market research databases.
- Utilized Azure services, such as Azure Blob Storage for data storage, to contribute to the management and transformation of large datasets while ensuring data accuracy.
- Employed Python and SQL to clean, preprocess, and manipulate data stored in Azure SQL Database, facilitating effective reporting and analysis.
- Developed interactive dashboards in Tableau to visualize key financial metrics, enhancing stakeholder engagement and understanding of financial health.
- Collaborated with team members across finance, risk, and compliance to gather requirements and define KPIs, ensuring alignment of analytical outputs with business objectives.
- Conducted exploratory data analysis (EDA) using Python to identify trends and outliers in financial data, providing actionable insights that informed client strategies.
- Presented analytical findings and recommendations to team members, contributing to inform decision-making and fostering a data-driven culture within the organization.

Technical Skills

- **Programming languages:** Python, PL/SQL
- **Databases:** Oracle, Teradata, SQL Server, PostgreSQL, MySQL, MongoDB (NoSQL), Cassandra (NoSQL)
- **Big Data Technologies**: HDFS, Apache Kafka, Hive, Yarn, Spark, PySpark, Airflow, Hadoop, Scala
- **Frameworks:** Spark/PySpark, NumPy, Pandas, PyTorch, TensorFlow, Scikit-Learn, Jupyter
- **Cloud Technologies:** AWS (EC2, Lambda, RDS, EMR, Kinesis, S3, SNS, Glue, Redshift, Cloud Watch, Athena, CloudFormation), **Azure** (Azure Blob Storage, Databricks, ADLS, Data Factory, Azure Synapse, Cosmos DB, EventHub, Functions, Azure DevOps)
- **Reporting Tools:** Tableau, Power BI
- **Methodologies:** Agile/Scrum, SDLC
- **Version Tools:** Git, GitHub, GitLab.

Certifications

- Earned Joy of Computing using Python certification – NPTEL
- Microsoft Certified: Azure Data Engineer Associate (DP-203)

Education

Duquesne University, Pittsburgh, PA
Master of Science in Computer Science Aug 2022 – May 2024

CMR Institute of Technology, India
Bachelor of Technology in Computer Science and Engineering Aug 2016 – Nov 2020

Projects

An Efficient Stenography Using Adaptive PVD Topic Modelling

- Installed a novel adaptive data-hiding method, achieving a 15% success rate in concealing secret data into grey-scale cover images.
- Operated LSB substitution and pixel-value differencing (PVD) techniques to ensure efficient data extraction and concealment.
- Employed Python, Django, HTML, and CSS for UI design, resulting in a seamless user experience.
- Developed a robust system capable of handling large volumes of data, achieving a processing speed of images per second.
- Enhanced security measures, resulting in a 90% reduction in data leakage incidents.
- Conducted testing and optimization, reducing system downtime by 80%.